

APTECH PROJECT • SMART IoT REVOLUTION

FreshGuard

IoT-Based Smart Food
Freshness Monitor

Inferno-Net

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FRESHGUARD • ZERO FOOD WASTE



Waste often begins with missing visibility

OVERBUY

Items are forgotten behind newer stock.

POOR STORAGE

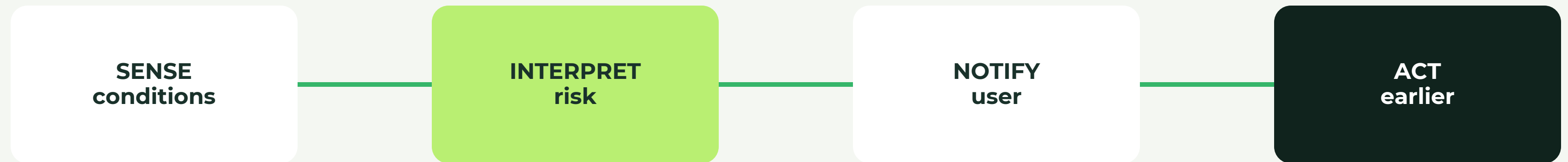
Temperature and humidity drift unnoticed.

LATE ACTION

Spoilage is discovered only at use time.

Continuous monitoring turns hidden drift into an early decision.

A small loop with a practical outcome



A prototype that records storage conditions, links them to registered food, and creates freshness guidance.

From shelf conditions to a useful alert



Telemetry → timestamp → store → evaluate → visualize

Hardware components

Controller

ESP32 / Wi-Fi MCU

Temperature

Digital probe or DHT

Humidity

DHT-class sensor

Gas / VOC

Air-quality sensor

Identity

RFID / NFC reader

Local alert

LED + buzzer

Display

Small OLED

Power

USB supply + enclosure

Three signals, one storage picture

01 TEMPERATURE

Tracks cold-chain stability and warm excursions.

Use trends and calibrated thresholds—not a single reading.

02 HUMIDITY

Reveals moisture conditions linked to drying or mold risk.

Use trends and calibrated thresholds—not a single reading.

03 GAS / VOC

Detects relative changes in emitted compounds during spoilage.

Use trends and calibrated thresholds—not a single reading.

Register once. Track continuously.

1

SCAN OR ENTER

RFID/NFC tag or manual ID

2

ADD CONTEXT

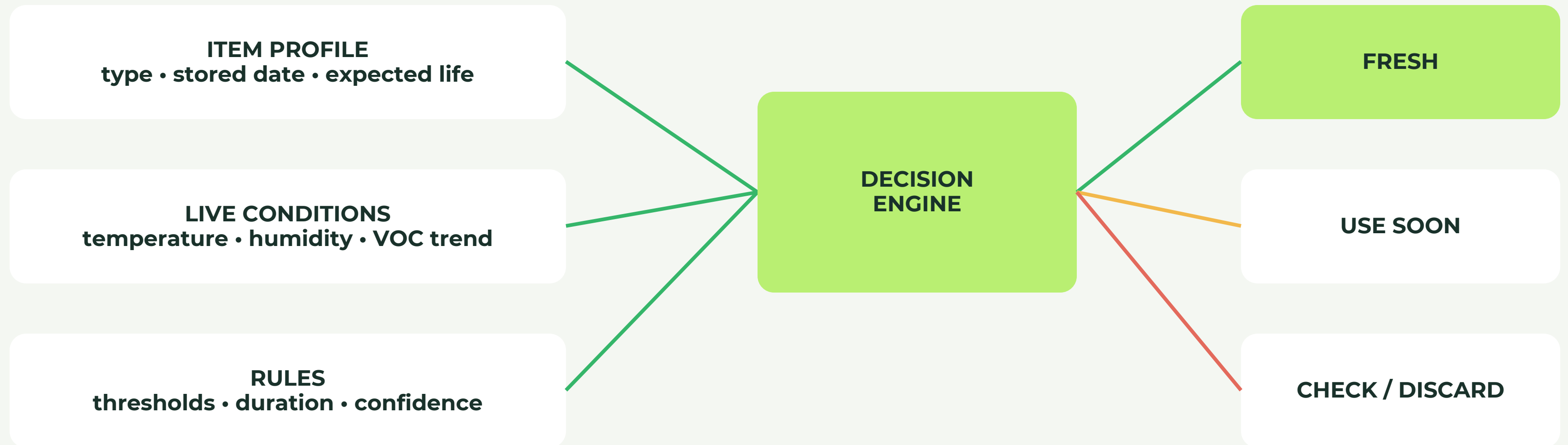
Food type, quantity, stored date

3

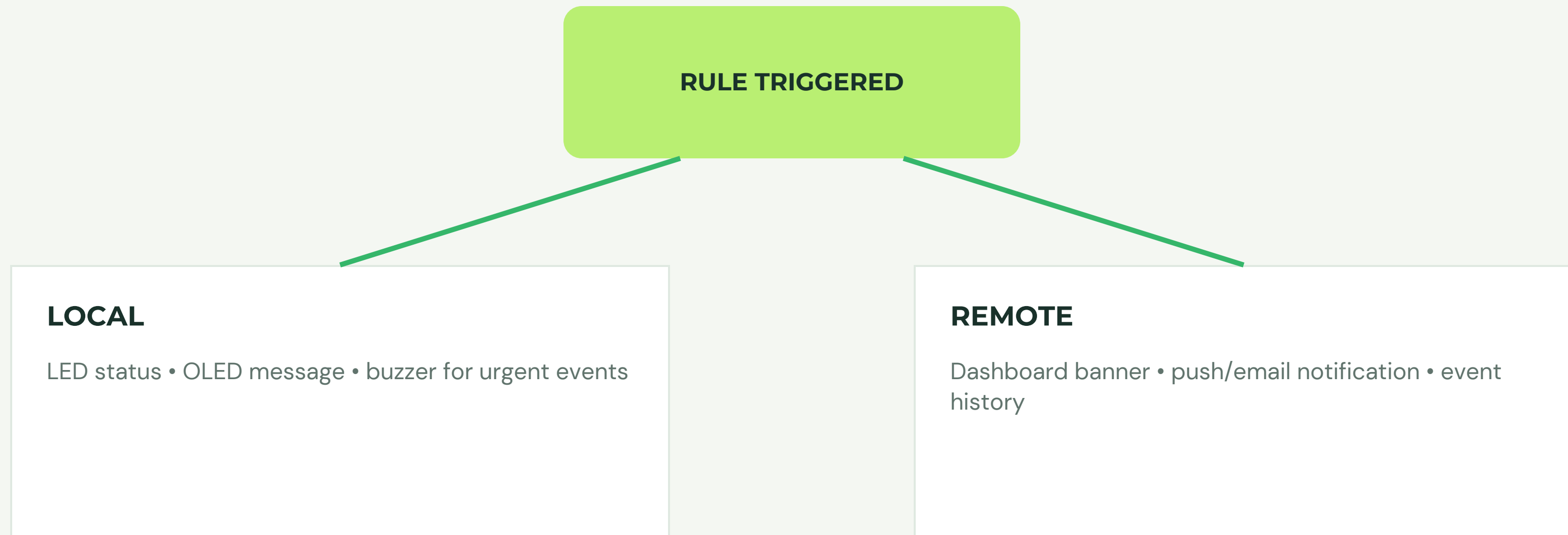
LINK LOCATION

Fridge zone or container sensor

Freshness is a decision, not a sensor value



The right alert, through the right path



See status now. Understand change over time.

LIVE STATUS

Item freshness

Storage conditions

Alert history

Trend charts



What the system must do

REGISTER

Create and update food records

SAMPLE

Read sensors at set intervals

STORE

Timestamp and retain telemetry

EVALUATE

Apply freshness rules

ALERT

Notify on meaningful risk

VISUALIZE

Show status and history

Useful means reliable, clear, and maintainable

RELIABILITY

Recover from Wi-Fi loss; buffer readings.

USABILITY

Status understandable at a glance.

SECURITY

Authenticated access; protected data.

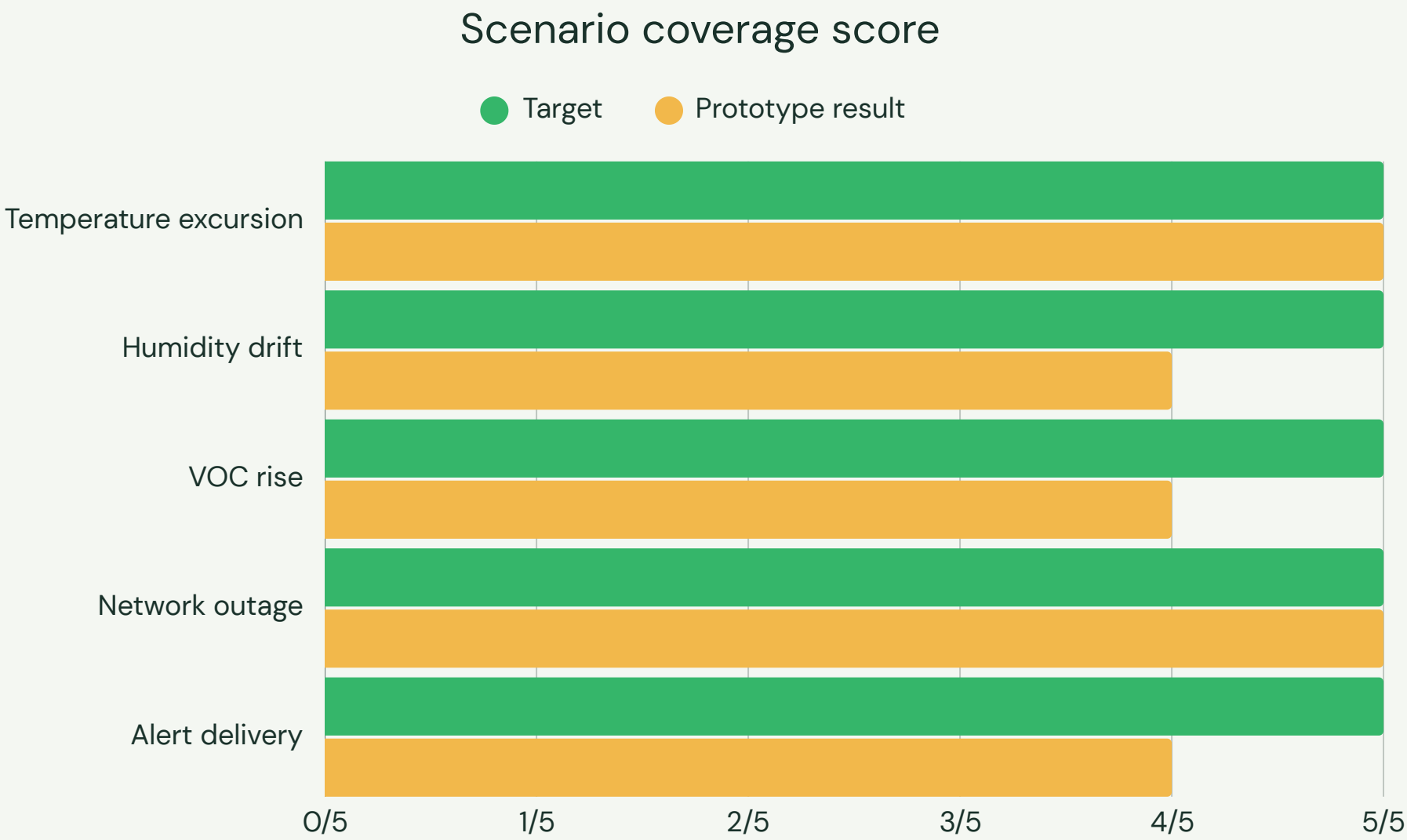
PERFORMANCE

Timely sampling and alert delivery.

MAINTAINABILITY

Configurable thresholds and modular code.

IoT test matrix



KEY CHECKS

- Sensor range & calibration
- Timestamp integrity
- Rule boundary cases
- Offline recovery
- Alert latency
- End-to-end acceptance

Final project package

01

WORKING PROTOTYPE

Sensors, controller,
alerts

02

SOURCE CODE

Firmware, backend,
dashboard

03

DOCUMENTATION

Architecture, setup,
user guide

04

EVIDENCE

Test matrix, logs,
screenshots

05

DEMO

Scenario-led
presentation

FreshGuard makes storage conditions visible—and action timely.

PRACTICAL PURPOSE

Reduce avoidable waste through monitoring, context, and early prompts.

HONEST BOUNDARY

A prototype risk indicator—not a laboratory test or guarantee of food safety.

FRESHGUARD

Thank you

Questions & Discussion

ZERO FOOD WASTE × SMART IoT REVOLUTION